

Canvas App — Flow Layout Verification Report

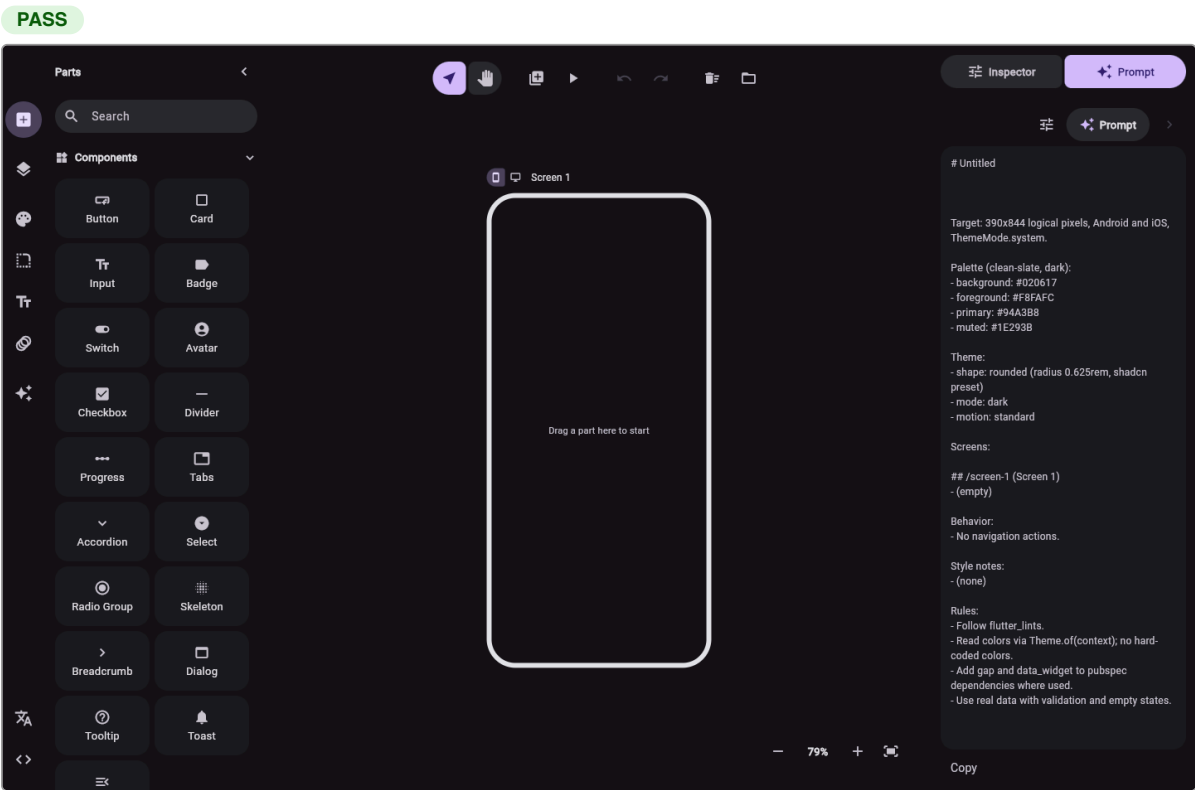
Batch P3: screen-as-Column flow layout, Row containers, persistence, device preview, tap fix · 3 research agents (Figma auto-layout / FlutterFlow+Draftbit / Framer+Webflow) + 6 build agents · Date: 2026-09-05 · Release bundle, 1440x900, driven live in Chrome
Gate: flutter analyze zero issues · flutter test 166/166 green · dart test (core) 30/30

What changed and why

Your feedback: parts landed at random absolute positions instead of stacking like written code. Fix (modeled on Figma auto-layout, FlutterFlow and Framer/Webflow stacks): the screen **is** a vertical Column (padding 16, gap 12); a Row part is an explicit horizontal container holding its own children; every drop inserts at an index (container highlight + 2px insertion line); the same tree renders in the editor, the preview, and the exported Flutter code (real Column/Row/SizeBox — verified by core tests).

Visual tour

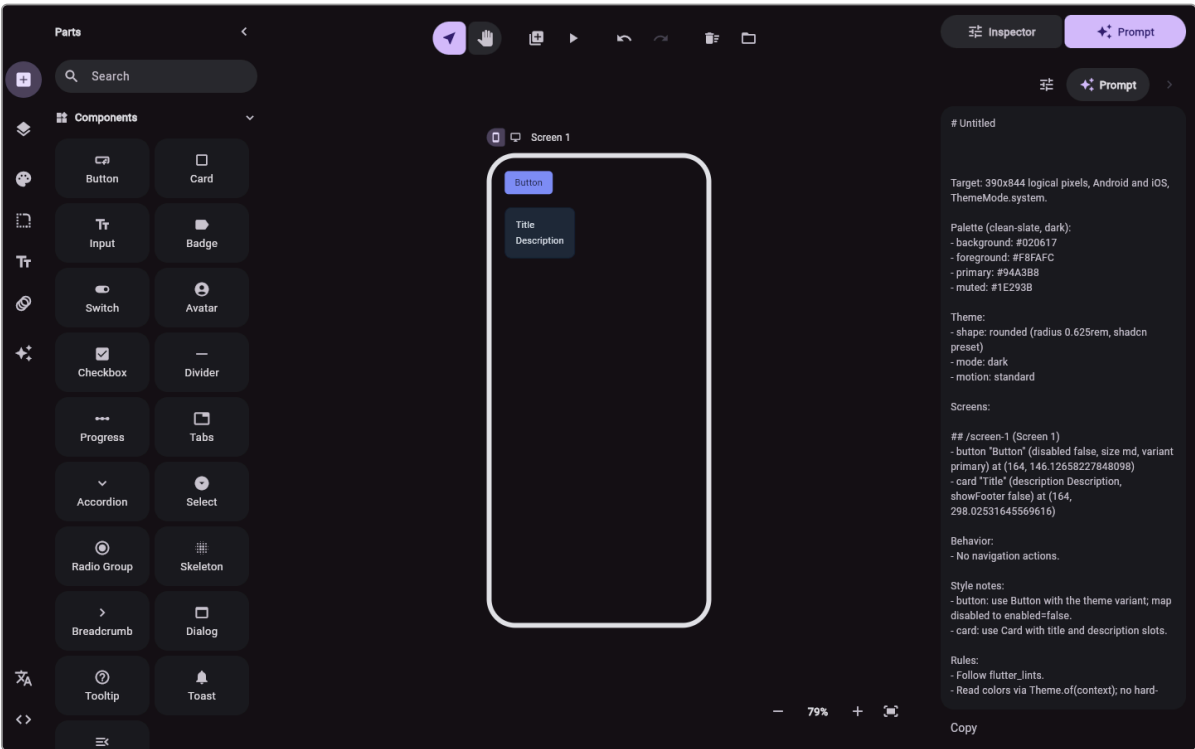
1 — Empty editor, 19-kind palette + Layout section



Palette grew to 19 content kinds plus a new Layout section holding the Row container tile. Shell unchanged; zero console errors.

2 — FLOW: dropped parts stack as a Column

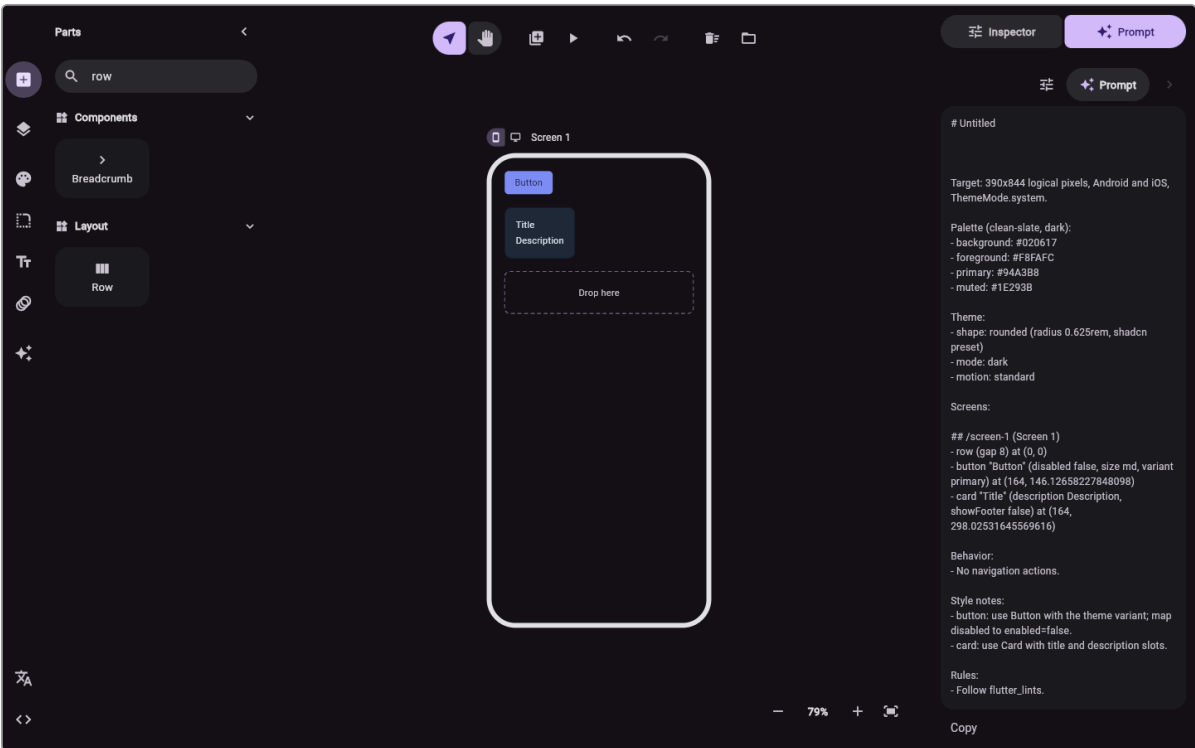
PASS



The core complaint of the last report, fixed: Button and Card dropped anywhere on the screen now reflow into a vertical column (Padding 16 > Column, gap 12) exactly as written Flutter code would. Absolute x/y placement is gone from the render path.

3 — Row container with 'Drop here' placeholder

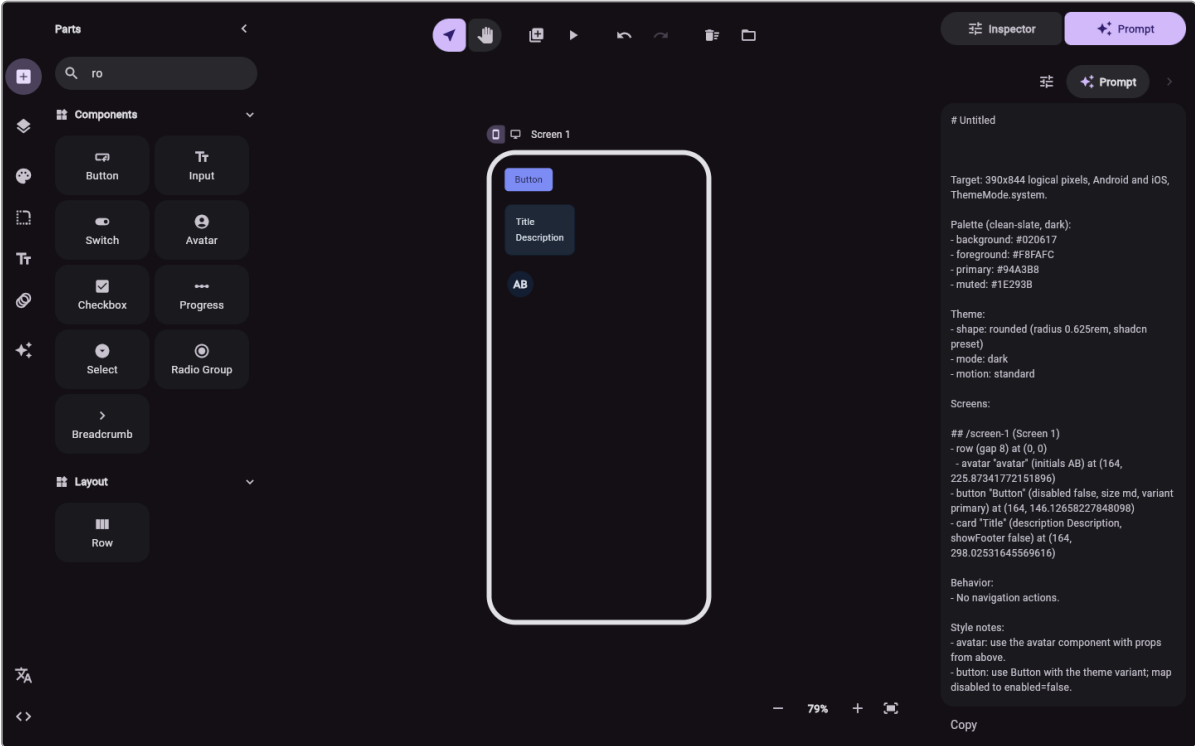
PASS



Dragging the Row tile creates a horizontal container rendered as a dashed drop target (min-height 64, always hit-testable). Prompt shows '- row (gap 8)' with children indented under it.

4 — Child dropped INSIDE the row

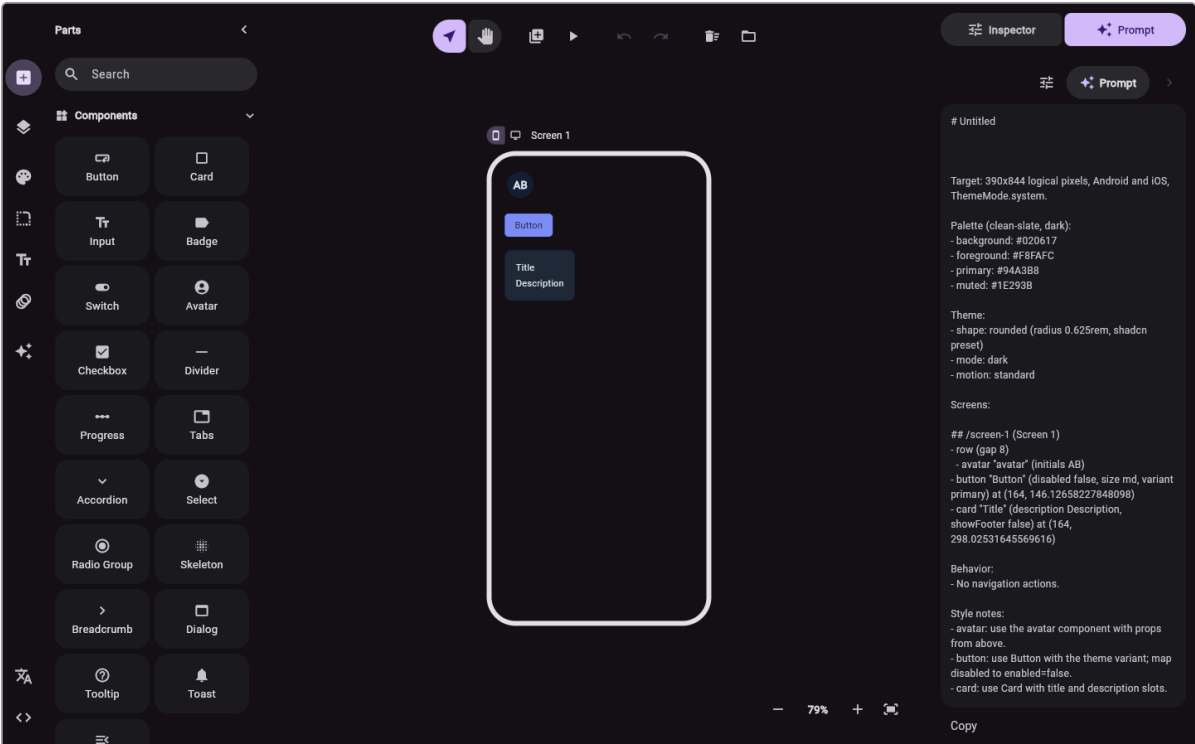
PASS



Dropping a part while hovering the row parents it (parentId) and renders side-by-side; insertion index line + container highlight guide the drop (shared insert/reorder path).

5 — Persistence: doc restored after reload

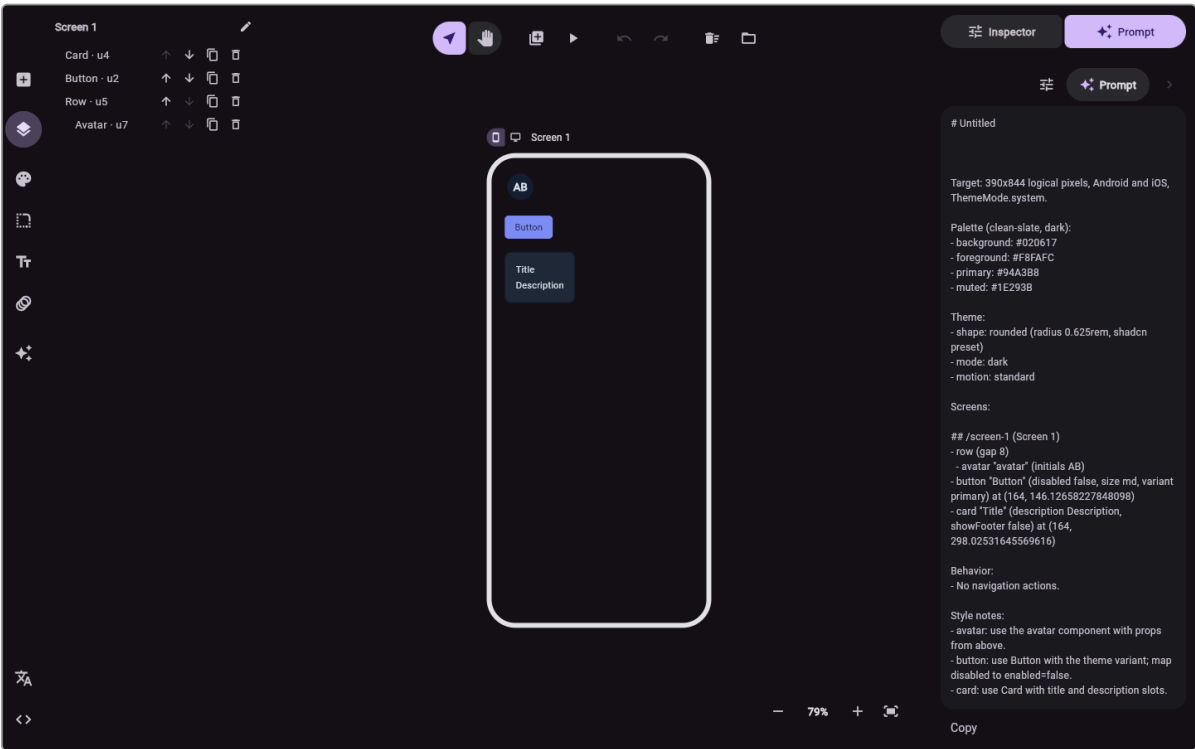
PASS



Previously dead autosave is now wired (SharedPreferencesPrefs + load()). After a full page reload the doc comes back, migrated to flow order (y-sorted roots). Legacy absolute docs auto-migrate on load.

6 — Layers as a tree (rows indented)

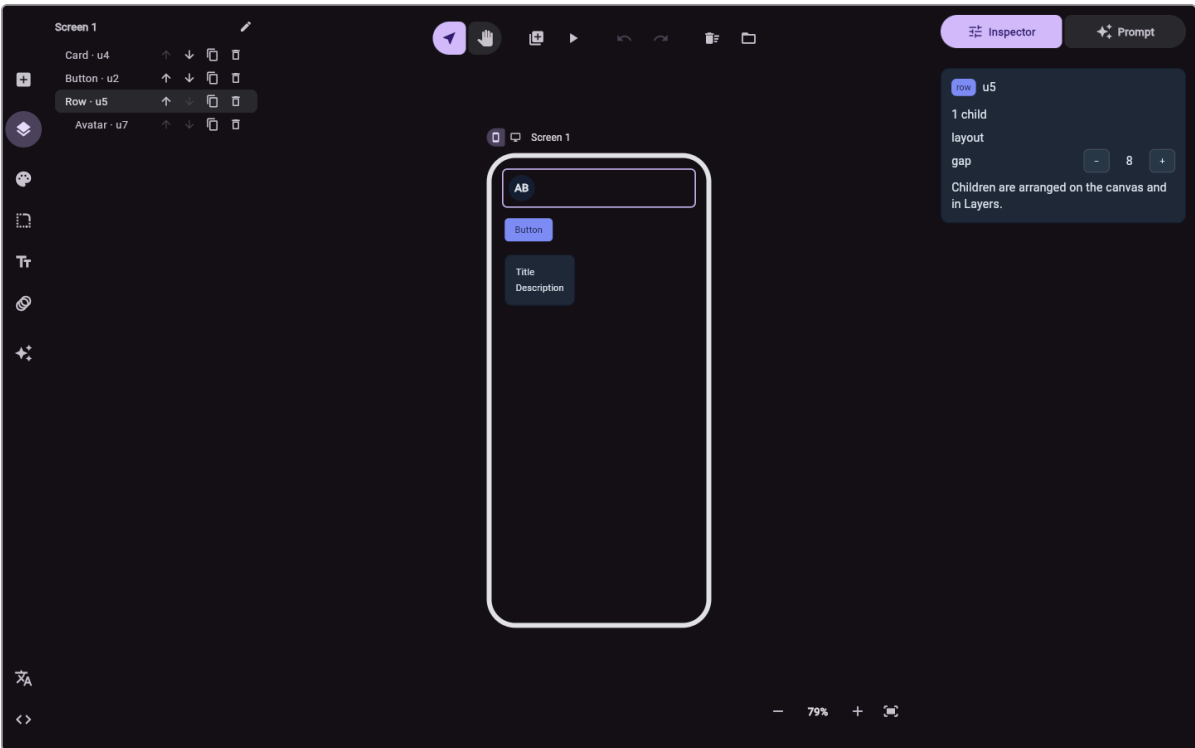
PASS



Layers now shows hierarchy: Row · u5 with Avatar · u7 indented beneath, reorder/rename/duplicate per row, screen rename pencil in the header.

7 — Row inspector: gap stepper

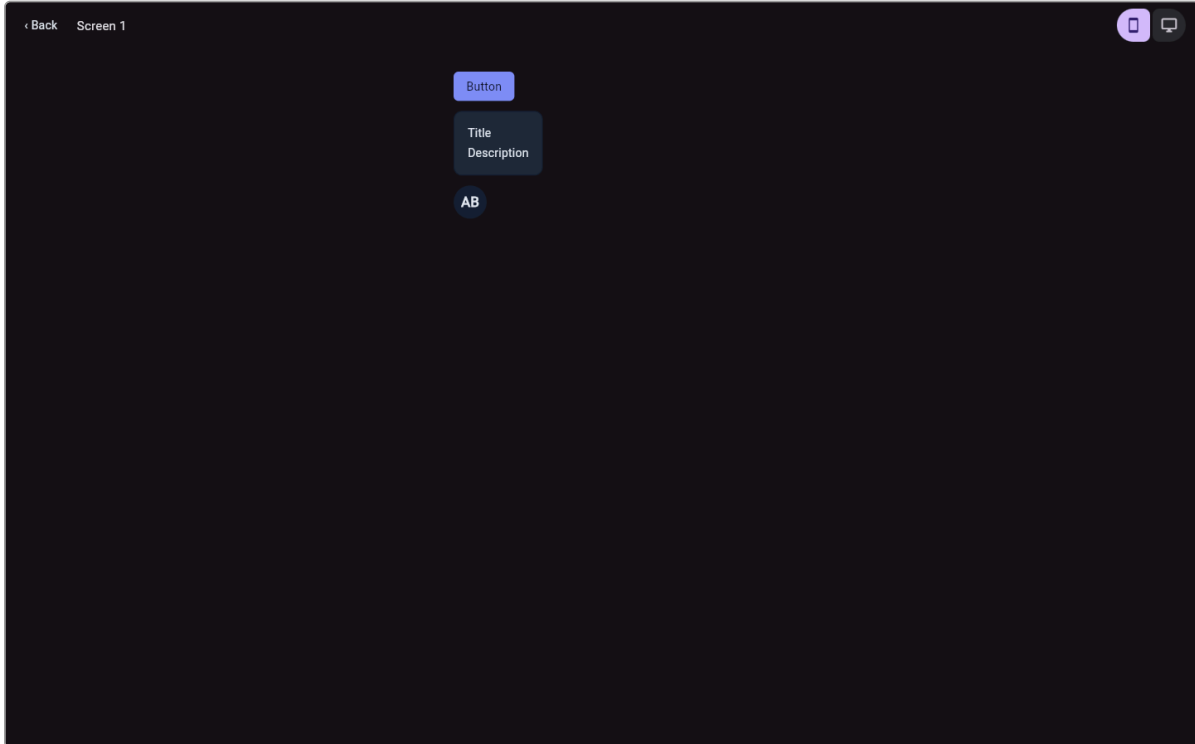
PASS



Selecting a row flips the Inspector to the container editor: row badge, child count, gap stepper (0–32 step 4) persisting through `store.patchNode` (undoable). Canvas re-renders the spacing live.

8 — Preview: mobile view (390 cap)

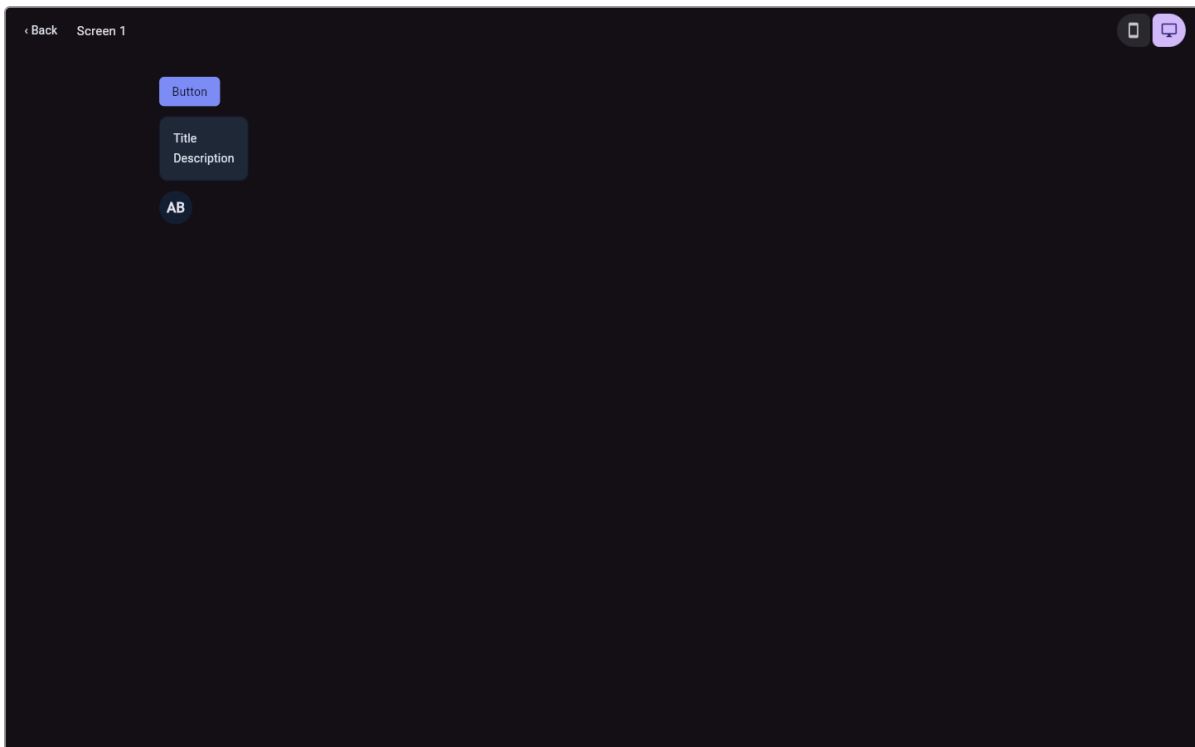
PASS



New segmented device toggle in the preview bar. Mobile caps the stage at 390px centered; same widget tree as the editor.

9 — Preview: web view (1100 fluid stage)

PASS



Desktop mode: frameless fluid stage (max 1100, centered) — the Webflow/Framer pattern. One tree, two widths; back-stack and interactions intact.

Bugs fixed this pass

Bug	Symptom	Root cause	Fix	Touched
Parts rendered at arbitrary absolute positions (the report-1 complaint)	Drops landed where the pointer was; nothing reflowed like real Flutter code.	Canvas was Stack+Positioned; flow requires ordered children of a Column/Row.	Doc model gained parentId/isRow/gap; screen renders Padding(16) > Column with SizedBox separators, rows are Row containers with children; drops insert at a computed index. Exporter + prompt emitter produce real Column/Row code from the same tree.	screen_doc/prompt_builder/codegen_single + phone_frame/design_canvas/store
Canvas taps dead on interactive parts (button/switch/...)	Tapping a dropped Button never selected it (ring/aside stayed put).	Kit widgets own inner GestureDetector.onTap recognizers that beat the node's selection recognizer in the gesture arena (proven with nested-GD probe: inner tap wins, outer 0).	Selection moved to an arena-exempt Listener.onPointerDown (opaque) on the leaf; pan-mode gated. 7/7 new regression tests.	phone_frame.dart
Search 'row' matched Breadcrumb	Filter matched kind substrings (b-ROW-crumb).	Known fuzzy-search artifact, cosmetic; left as-is this batch.	Noted for a later pass (match on label only).	—

Console-error tally

0 errors across empty load, 6+ drags (column + row drops), layers selection, row gap edits, persistence reload, and both preview modes.

Open items

- Row gap editing exists; row main/cross alignment + child flex/expand are v2 (research-backed defaults shipped: start/center).
- Light-theme canvas-well tokens and single motion value unchanged (documented).
- Search matches kind substrings (cosmetic).
- HANDOFF leftovers: keyboard-shortcut help overlay, more catalog kinds, per-breakpoint overrides (v2).